

Assignment 3

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Econometrics I
NYU Stern

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Instructions. This assignment covers endogeneity, simultaneity, and instrumental variables (Sessions 5–6). You may use R (with `fixest`; `feols(y ~ x | p ~ z)` estimates 2SLS) or Python (with `pyfixest`). Turn in (1) a PDF presenting your results and showing your work, and (2) your code. As always: derive with the operator rules, then simulate to check yourself.

1 Why you cannot regress quantity on price

A grocery chain sets weekly prices for a snack product in 500 store-weeks. Demand and pricing are:

$$\begin{aligned} q_i &= 8 - 2 \ln p_i^* + \xi_i && \text{(demand, in logs: elasticity } -2) \\ \ln p_i^* &= 1 + 0.5 c_i + 0.3 \xi_i && \text{(pricing: costs, and a response to demand)} \end{aligned}$$

where $\xi_i \sim \mathcal{N}(0, 0.5^2)$ is a demand shock the chain observes but you do not, and $c_i \sim \mathcal{N}(0, 0.5^2)$ is a wholesale cost shifter.

- Before simulating: use the operator rules to derive $\text{Cov}(\ln p^*, \xi)$ and the sign of the bias in the OLS regression of q on $\ln p^*$. Will OLS make demand look too elastic or too inelastic? Give the economic story in one sentence.
- Simulate 500 observations. Estimate the elasticity by OLS and compare to your answer in (a).
- Now use the cost shifter c as an instrument for $\ln p^*$ (`feols(q ~ 1 | lnp ~ c)`). Compare the estimate to -2 .

- (d) State the two IV conditions (relevance, exclusion) in the language of this problem, and say how you would defend each to a skeptical seminar audience. One of them is checkable in data — which one, and how?
- (e) A colleague proposes instrumenting price with *the same product’s price at other stores in the chain*. Under what assumption is that valid, and what plausible feature of grocery chains breaks it?

2 The randomized experiment that wasn’t enough

Based on Jeziorski, Leng & Seiler (2025), “Causal Inference with Endogenous Price Response” (working paper; CEPR DP 20252) — see the DAG from Session 5.

An online marketplace randomly assigns a product-page upgrade ($T_i \in \{0, 1\}$, probability 0.5) to 5,000 sellers. The upgrade shifts demand directly, but sellers also *reprice* in response to the upgrade and to demand conditions:

$$\begin{aligned} p_i &= 1 + 0.3T_i + 0.4\xi_i + 0.5c_i \\ q_i &= 2 - 2p_i + 1 \cdot T_i + \xi_i \end{aligned}$$

with $\xi_i, c_i \sim \mathcal{N}(0, 1)$ independent; c_i is a seller-level cost shifter. The **direct effect** of the upgrade — the demand shift holding price constant — is 1.

- (a) Simulate the data. Regress q on T alone. What number do you get, and *of what is it an unbiased estimate?* (Hint: substitute the price equation into the demand equation.) When would a manager want exactly this number?
- (b) Now “control for price”: regress q on T and p . You should find an estimate strictly between (a) and the direct effect of 1. Explain, using the DAG from lecture, why this regression is biased for the direct effect *even though T is randomized*. Which arrow does the damage?
- (c) Run the diagnostic from lecture: regress p on T . What do you find, and what does it tell you?
- (d) Recover the direct effect by instrumenting price with the cost shifter: `feols(q ~ T | p ~ c)`. Compare to 1.
- (e) Managerial summary, three sentences max: your product team says “the upgrade only moved sales by [your answer to (a)], so it’s not worth the engineering cost.” What do you tell them, and which of your four numbers belongs in which business decision?
- (f) Re-run (a)–(d) with the repricing channel shut off ($p_i = 1 + 0.4\xi_i + 0.5c_i$). Which estimators now agree, and why does the diagnostic in (c) predict this?

3 The Wald estimator, by hand

Consider the just-identified model $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ with a binary instrument z_i satisfying $\text{Cov}(z, \varepsilon) = 0$ and $\text{Cov}(z, x) \neq 0$.

- (a) Starting from the model, take the covariance of both sides with z (operator rules!) and solve for β_1 . You have just derived $\beta_1 = \text{Cov}(z, y) / \text{Cov}(z, x)$.
- (b) Show that this equals the *Wald estimator* $(\mathbb{E}[y|z = 1] - \mathbb{E}[y|z = 0]) / (\mathbb{E}[x|z = 1] - \mathbb{E}[x|z = 0])$.
- (c) In one sentence: why does dividing the “reduced form” by the “first stage” undo the attenuation that a weak first stage would otherwise cause — and what is the price you pay when the first stage is very weak?